Because a black hole has no “hair,” all one can discover, by examining the hole from outside, are the total amounts of mass, angular momentum, and charge that entered the atmosphere.

Ask how many ways those amounts of mass, angular momentum, and charge *could* have been injected into the hole’s hot atmosphere. This question is analogous to asking how many ways the child’s toys *could* have been distributed over the tiles in the playroom of Box 12.3, and correspondingly, the logarithm of the number of ways to inject must be the increase in the atmosphere’s entropy, as described by the standard laws of thermodynamics. By a fairly simple calculation, Zurek and I were able to show that this increase in thermodynamic entropy is precisely equal to ¼ times the increase in the horizon’s area, divided by the Planck–Wheeler area; that is, it is precisely the increase in the horizon’s area in disguise, the same disguise that Hawking inferred, in 1974, from the mathematical similarity of the laws of black-hole mechanics and the laws of thermodynamics.

The outcome of this thought experiment can be expressed succinctly as follows: *A black hole’s entropy is the logarithm of the number of ways that the hole could have been made.* This means that there are $10^{4.6 \times 10^{78}}$ different ways to make a 10-solar-mass black hole whose entropy is 4.6×10^{78}. This explanation of the entropy was originally conjectured by Bekenstein in 1972, and a highly abstract proof was given by Hawking and his former student, Gary Gibbons, in 1977.

The thought experiment also shows the second law of thermodynamics in action. The energy, angular momentum, and charge that one throws into the hole’s atmosphere can have any form at all; for example, they might be the roomful of air wrapped up in a bag, which we met earlier in this chapter while puzzling over the second law. When the bag is thrown into the hole’s atmosphere, the entropy of the external Universe is reduced by the amount of entropy (randomness) in the bag. However, the entropy of the hole’s atmosphere, and thence of the hole, goes up by more than the bag’s entropy, so the total entropy of hole plus external Universe goes up. The second law of thermodynamics is obeyed.

Similarly, it turns out, when the black hole evaporates some particles, its own surface area and entropy typically go down; but the particles get